

Q1. Covariance and Correlation

Children of three age groups are asked to indicate their preference for three photographs of adults. The data are given below:

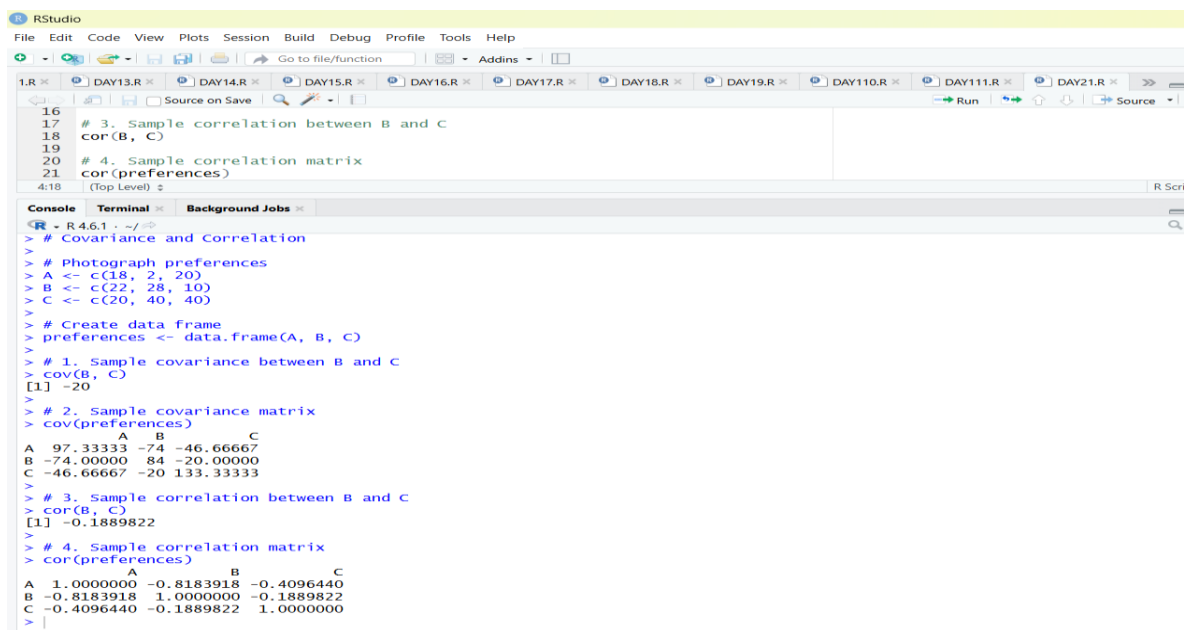
Age of Child	Photograph A	Photograph B	Photograph C
5–6 years	18	22	20
7–8 years	2	28	40
9–10 years	20	10	40

1. Use `cov()` to calculate the sample covariance between B and C.
2. Use `cov()` to calculate the sample covariance matrix for A, B, and C.
3. Use `cor()` to calculate the sample correlation between B and C.
4. Use `cor()` to calculate the sample correlation matrix for A, B, and C.

R CODE:

```
A <- c(18, 2, 20)
B <- c(22, 28, 10)
C <- c(20, 40, 40)
preferences <- data.frame(A, B, C)
cov(B, C)
cov(preferences)
cor(B, C)
cor(preferences)
```

OUTPUT:



```
RStudio
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1.R DAY13.R DAY14.R DAY15.R DAY16.R DAY17.R DAY18.R DAY19.R DAY110.R DAY111.R DAY21.R
16
17 # 3. Sample correlation between B and C
18 cor(B, C)
19
20 # 4. Sample correlation matrix
21 cor(preferences)
4:18 (Top Level)

Console Terminal Background Jobs
R - R4.6.1 - /
> # Covariance and Correlation
>
> # Photograph preferences
> A <- c(18, 2, 20)
> B <- c(22, 28, 10)
> C <- c(20, 40, 40)
>
> # Create data frame
> preferences <- data.frame(A, B, C)
>
> # 1. Sample covariance between B and C
> cov(B, C)
[1] -20
>
> # 2. Sample covariance matrix
> cov(preferences)
      A      B      C
A 97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
>
> # 3. Sample correlation between B and C
> cor(B, C)
[1] -0.1889822
>
> # 4. Sample correlation matrix
> cor(preferences)
      A      B      C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918 1.0000000 -0.1889822
C -0.4096440 -0.1889822 1.0000000
>
```

Q2. All Electronics Price Data – Binning and Histogram

Question:

Imagine that you have selected data from the **All Electronics data warehouse** for analysis. The data set will be huge. The following data are a list of **All Electronics prices for commonly sold items**, rounded to the nearest dollar. The numbers have been sorted:

1, 1, 5, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30.

Perform the following:

- (i) Partition the dataset using an **equal-frequency partitioning method** with **bin size equal to 3**.
- (ii) Apply **data smoothing using bin means and bin boundaries**.
- (iii) Plot a **histogram** for the above frequency distribution.

R CODE:

```
price <- c(1,1,5,5,5,5,5,8,8,10,10,10,10,12,
          14,14,14,15,15,15,15,15,15,
          18,18,18,18,18,18,18,18,
          20,20,20,20,20,20,20,
          21,21,21,21,
          25,25,25,25,25,
          28,28,30,30,30)

price <- sort(price)

bin_size <- 3

bins <- split(price, ceiling(seq_along(price) / bin_size))

print("Bins:")

print(bins)

bin_means <- sapply(bins, mean)

print("Bin Means:")

print(bin_means)

boundary_smooth <- unlist(lapply(bins, function(x) {
  lower <- min(x)
  upper <- max(x)
  ifelse(abs(x-lower) <= abs(x-upper), lower, upper)
}))
```

```
print("Bin Boundary Smoothing:")

print(boundary_smooth)

hist(price,

      main = "Histogram of All Electronics Prices",

      xlab = "Price",

      ylab = "Frequency")
```

OUTPUT:

```

Console Terminal Background Jobs
R - R4.6.1 - ~/
> price <- c(1,1,5,5,5,5,5,8,8,10,10,10,10,12,
+           14,14,14,15,15,15,15,15,15,
+           18,18,18,18,18,18,18,18,
+           20,20,20,20,20,20,20,
+           21,21,21,21,
+           25,25,25,25,25,
+           28,28,30,30,30)
> price <- sort(price)
> bin_size <- 3
> bins <- split(price, ceiling(seq_along(price) / bin_size))
> print("Bins:")
[1] "Bins:"
> print(bins)
$`1`
[1] 1 1 5
$`2`
[1] 5 5 5
$`3`
[1] 5 8 8
$`4`
[1] 10 10 10
$`5`
[1] 10 12 14
$`6`
[1] 14 14 15
$`7`
[1] 15 15 15
$`8`
[1] 15 15 18
$`9`
[1] 18 18 18
$`10`
[1] 18 18 18
$`11`
[1] 18 20 20
$`12`
[1] 20 20 20
$`13`
[1] 20 20 21
$`14`
[1] 21 21 21
$`15`
[1] 25 25 25
$`16`
[1] 25 25 28
$`17`
[1] 28 30 30
$`18`
[1] 30

> # Bin mean
> bin_means <- sapply(bins, mean)
> print("Bin Means:")
[1] "Bin Means:"
> print(bin_means)
      1      2      3      4      5      6      7      8      9     10
11 2.333333 5.000000 7.000000 10.000000 12.000000 14.333333 15.000000 16.000000 18.000000 18.000000
12 19.333333 20.000000
13 20.333333 21.000000 25.000000 26.000000 29.333333 30.000000
14
15
16
17
18

> # Bin boundary smoothing
> boundary_smooth <- unlist(lapply(bins, function(x) {
+   lower <- min(x)
+   upper <- max(x)
+   ifelse(abs(x-lower) <= abs(x-upper), lower, upper)
+ }))
> print("Bin Boundary Smoothing:")
[1] "Bin Boundary Smoothing:"
> print(boundary_smooth)
 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103

```



```

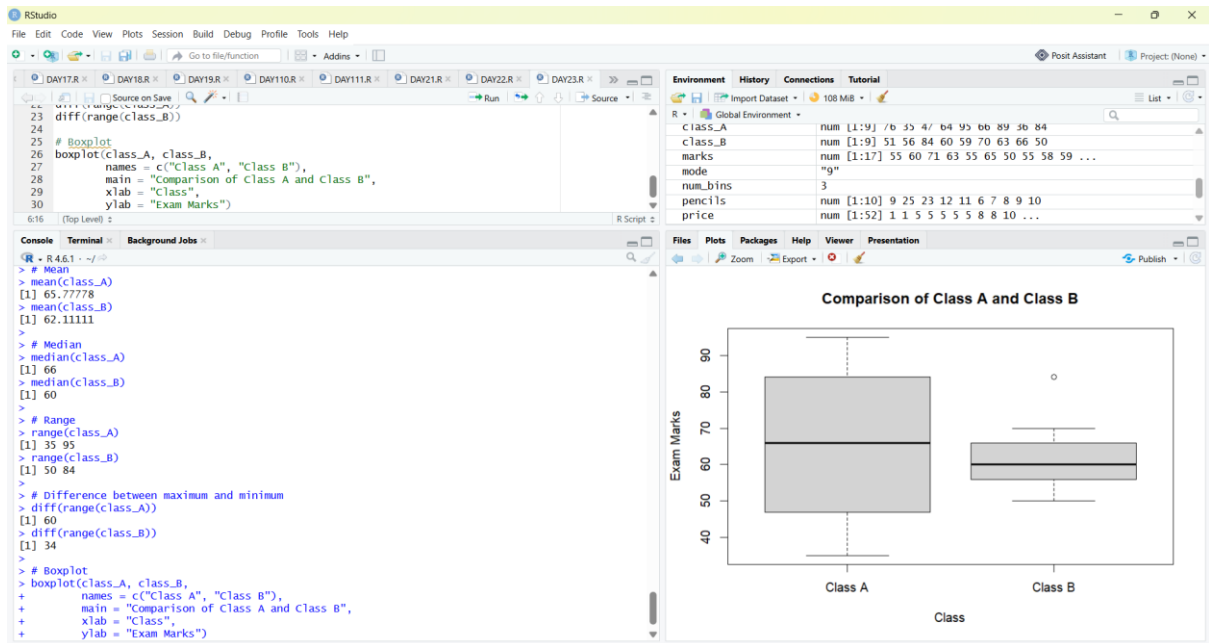
main = "Comparison of Class A and Class B",

xlab = "Class",

ylab = "Exam Marks")

```

OUTPUT:



Q4. Let us consider one example to make the calculation method clear. Assume that the minimum and maximum values for the feature F are \$50,000 and \$100,000 correspondingly.

It needs to range F from 0 to 1. In accordance with min-max normalization, $v = \$80$,

b) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

R CODE:

```
x <- c(200, 300, 400, 600, 1000)
```

```
min_max_norm <- (x - min(x)) / (max(x) - min(x))
```

```
print("Min-Max Normalization:")
```

```
print(min_max_norm)
```

```
z_score_norm <- (x - mean(x)) / sd(x)
```

```
print("Z-Score Normalization:")
```

```
print(z_score_norm)
```

OUTPUT:

```
RStudio
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DAY13.R DAY14.R DAY15.R DAY16.R DAY17.R DAY18.R DAY19.R DAY110.R DAY111.R DAY21.R DAY22.R DAY23.R DAY24.R
1 # Q4 - Normalization
2
3 # Data
4 x <- c(200, 300, 400, 600, 1000)
5
6 # (a) Min-Max Normalization
7 min_max_norm <- (x - min(x)) / (max(x) - min(x))
8
9 print("Min-Max Normalization:")
10 print(min_max_norm)
11
12
13 # (b) Z-Score Normalization
14 z_score_norm <- (x - mean(x)) / sd(x)
15
16 print("Z-Score Normalization:")
17 print(z_score_norm)
18
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27
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83
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98
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100
```

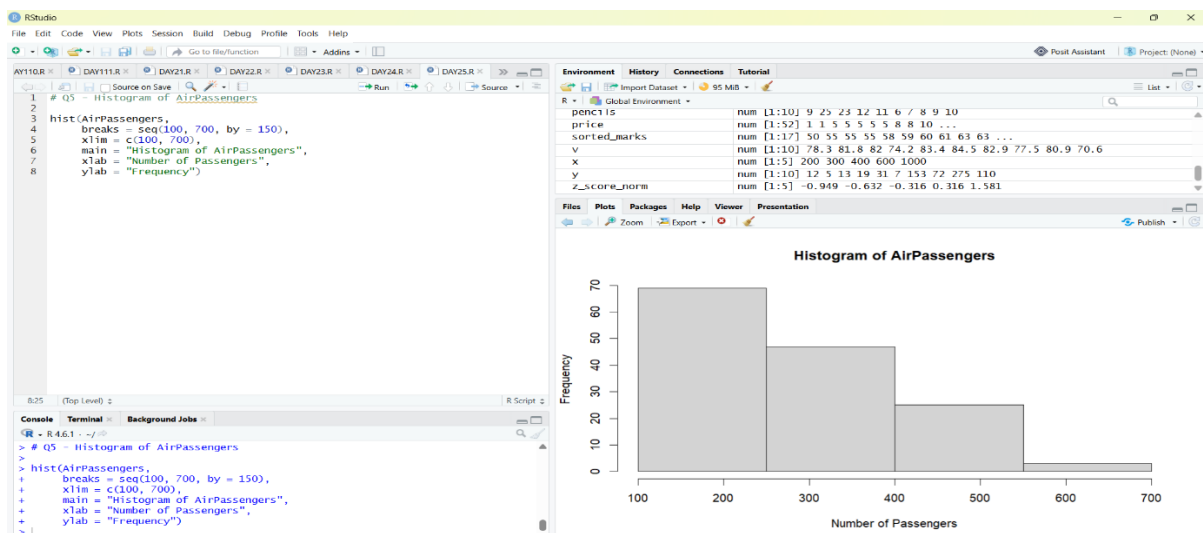
```
R - R 4.6.1 - ~/
> # Data
> x <- c(200, 300, 400, 600, 1000)
> # (a) Min-Max Normalization
> min_max_norm <- (x - min(x)) / (max(x) - min(x))
> print("Min-Max Normalization:")
[1] "Min-Max Normalization:"
> print(min_max_norm)
[1] 0.000 0.125 0.250 0.500 1.000
>
> # (b) Z-Score Normalization
> z_score_norm <- (x - mean(x)) / sd(x)
> print("Z-Score Normalization:")
[1] "Z-Score Normalization:"
> print(z_score_norm)
[1] -0.9486833 -0.6324555 -0.3162278 0.3162278 1.5811388
```

Q5. Make a histogram for the “AirPassengers” dataset, start at 100 on the x-axis, and from values 200 to 700, make the bins 150 wide

R CODE:

```
hist(AirPassengers,
     breaks = seq(100, 700, by = 150),
     xlim = c(100, 700),
     main = "Histogram of AirPassengers",
     xlab = "Number of Passengers",
     ylab = "Frequency")
```

OUTPUT:



Q6. Obtain Multiple Lines in Line Chart using a single Plot Function in R. Use attributes “mpg” and “qsec” of the dataset “mtcars”

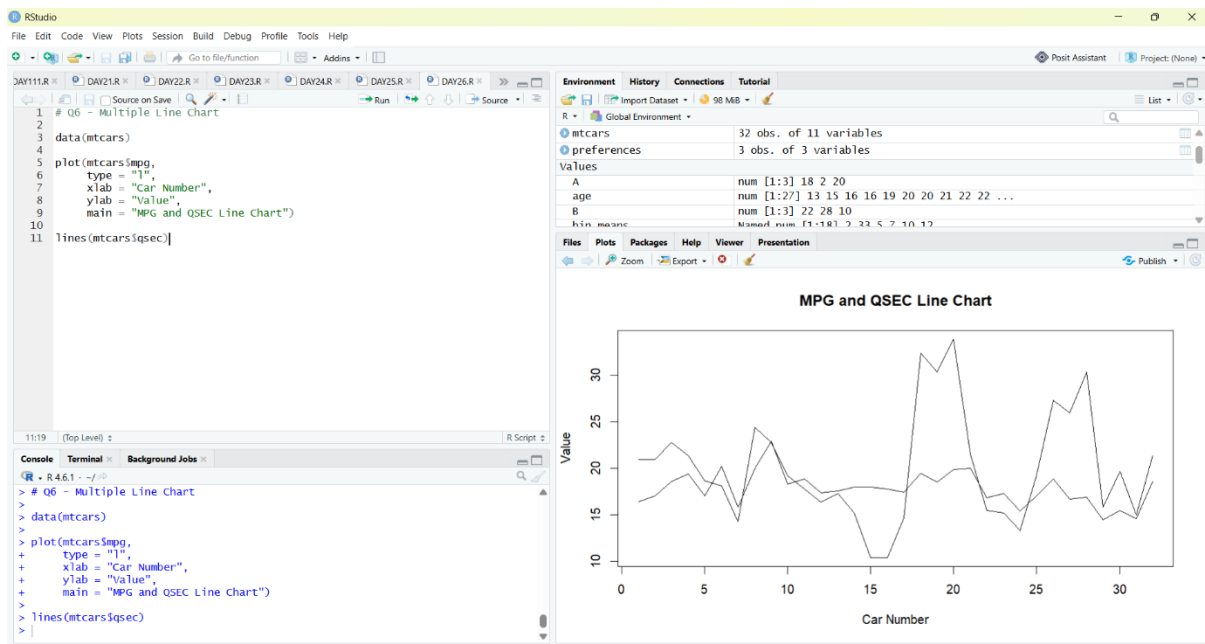
R CODE:

```
data(mtcars)

plot(mtcars$mpg,
     type = "l",
     xlab = "Car Number",
     ylab = "Value",
     main = "MPG and QSEC Line Chart")

lines(mtcars$qsec)
```

OUTPUT:



Q7. Download the Dataset “water” From R dataset Link. Find out whether there is a linear relation between attributes “mortality” and “hardness” by plot function. Fit the Data into the Linear Regression model. Predict the mortality for the hardness=88.

R CODE:

```
library(MASS)

data(water)

water

plot(water$hardness,
```

```

water$mortality,

main = "Mortality vs Hardness",

xlab = "Water Hardness",

ylab = "Mortality",

pch = 19)

model <- lm(mortality ~ hardness, data = water)

summary(model)

abline(model)

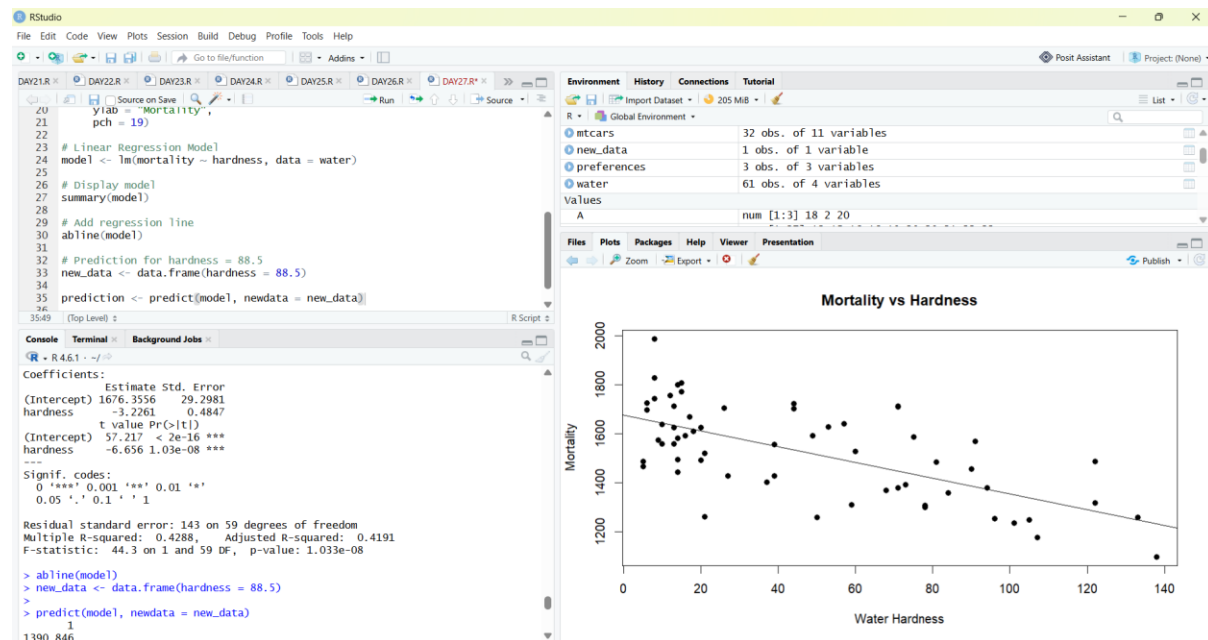
new_data <- data.frame(hardness = 88.5)

prediction <- predict(model, newdata = new_data)

print(prediction)

```

OUTPUT:



8. Create a Boxplot graph for the relation between "mpg" (miles per gallon) and "cyl" (number of Cylinders) for the dataset "mtcars" available in R Environment.

R CODE:

```

data(mtcars)

boxplot(mpg ~ cyl,

data = mtcars,

main = "MPG vs Number of Cylinders",

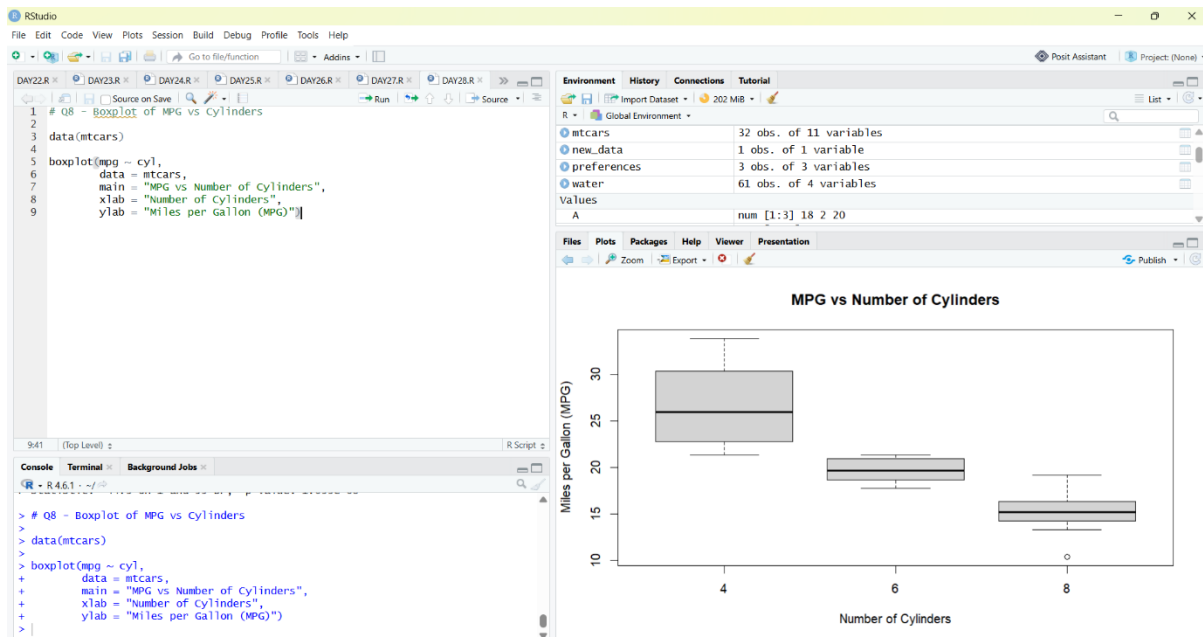
```

```

xlab = "Number of Cylinders",
ylab = "Miles per Gallon (MPG)"

```

OUTPUT:



9. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? Give suitable example using Boxplot visualization technique.

R CODE:

```

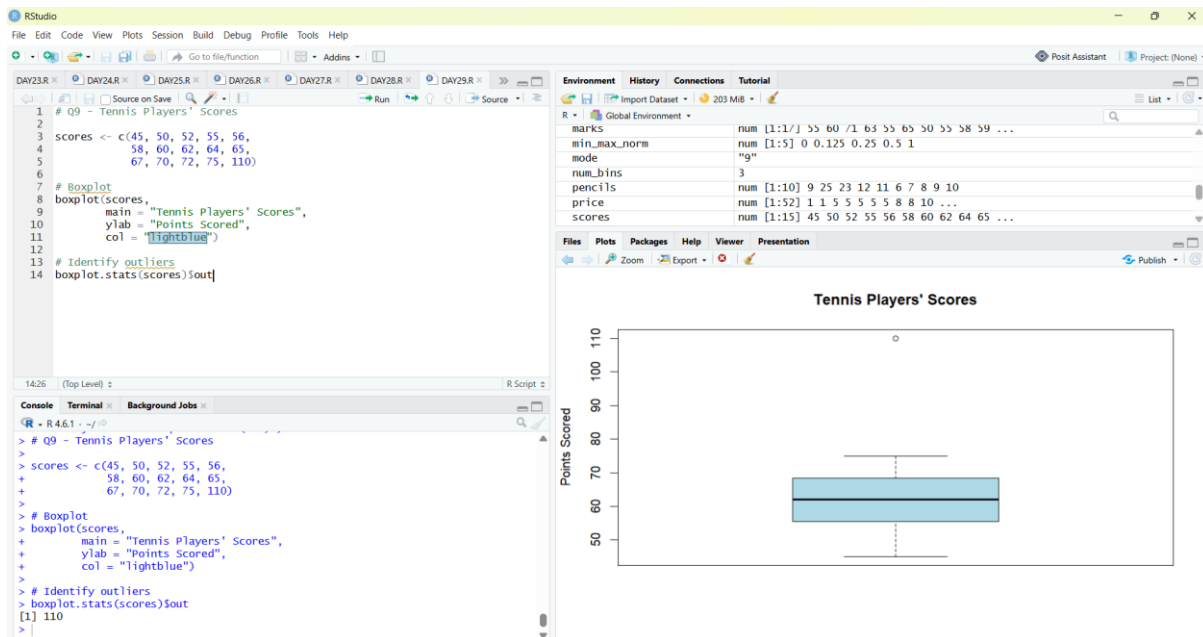
scores <- c(45, 50, 52, 55, 56,
            58, 60, 62, 64, 65,
            67, 70, 72, 75, 110)

boxplot(scores,
        main = "Tennis Players' Scores",
        ylab = "Points Scored",
        col = "lightblue")

boxplot.stats(scores)$out

```

OUTPUT:



10. Implement using R language in which age group of people are affected by blood pressure based on the diabetes dataset show it using scatterplot and bar chart (that is BloodPressure vs Age using dataset "diabetes.csv")

R CODE:

```
diabetes <- read.csv("diabetes.csv")
```

```
head(diabetes)
```

```
names(diabetes)
```

```
plot(diabetes$Age,
```

```
      diabetes$BloodPressure,
```

```
      main = "Blood Pressure vs Age",
```

```
      xlab = "Age",
```

```
      ylab = "Blood Pressure",
```

```
      pch = 19)
```

```
diabetes$AgeGroup <- cut(diabetes$Age,
```

```
                        breaks = c(0,20,30,40,50,60,100),
```

```
                        labels = c("0-20","21-30","31-40",
```

```
                                "41-50","51-60","60+"))
```

```
avg_bp <- aggregate(BloodPressure ~ AgeGroup,
```

```
                    data = diabetes,
```

```
                    FUN = mean)
```

```
# Display average BP
```

```
print(avg_bp)
```

```
barplot(avg_bp$BloodPressure,
```

```
names.arg = avg_bp$AgeGroup,
```

```
main = "Average Blood Pressure by Age Group",
```

```
xlab = "Age Group",
```

```
ylab = "Average Blood Pressure")
```

OUTPUT:

